Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-2(P1) | Date: 29-03-26
Time: 3 Hrs | 2020_P1 | Max. Marks: 198

KEY SHEET
MATHS

1	**C**	2	**B**	3	**C**	4	**C**	5	**D**
6	**A**	7	**CD**	8	**ABD**	9	**ABC**	10	**ABC**
11	**AD**	12	**AC**	13	**2**	14	**2**	15	**90**
16	**432**	17	**0**	18	**1**				

PHYSICS

19	**C**	20	**A**	21	**D**	22	**C**	23	**D**
24	**D**	25	**AD**	26	**BD**	27	**AB**	28	**AB**
29	**ABCD**	30	**AD**	31	**7**	32	**2**	33	**1**
34	**0.12 to 0.14**	35	**8**	36	**1.50 to 1.65**				

CHEMISTRY

37	**A**	38	**C**	39	**B**	40	**C**	41	**D**
42	**C**	43	**ABC**	44	**AB**	45	**AB**	46	**ABCD**
47	**BC**	48	**ABD**	49	**5**	50	**2**	51	**2**
52	**3**	53	**40.00 to 40.20**	54	**18.90 to 18.92**				

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SOLUTIONS
MATHS

1. $\tan\theta_1 + \tan\theta_2 = -a$

$\tan\theta_1 \cdot \tan\theta_2 = b \Rightarrow$

$\cot\theta_1 + \cot\theta_2 = -c$

$\cot\theta_1 \cdot \cot\theta_2 = d$

$\frac{1}{b} = d \Rightarrow bd = 1$

$\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2}$

$\theta_1 + \theta_2 = \pi/4 \Rightarrow \tan\theta_1 + \tan\theta_2 + \tan\theta_1 \tan\theta_2 = 1$

$-a + b = 1 \quad \rightarrow (1)$

$P(1)Q(1) = 16$

$(1 + a + b)(1 + c + d) = 16$

$(2b)\left(1 + \frac{a}{b} + \frac{1}{b}\right) = 16 \Rightarrow (2b)\left(\frac{b + a + 1}{b}\right) = 16$

$\frac{(2b)(2b)}{b} = 16 \Rightarrow \frac{4b^2}{b} = 16$

$\Rightarrow b = 4$

$\Rightarrow b = 1 + a$

$\Rightarrow a = 3$

$\frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2} = -c \Rightarrow 3 = c \cdot 4$

$\frac{\tan\theta_2 + \tan\theta_1}{\tan\theta_1 \tan\theta_2} = -c \Rightarrow c = \frac{3}{4}$

$-a = -cb$

$a = cb \Rightarrow c = \frac{a}{b}$

$c = ad$

$\Rightarrow \frac{a + c}{1 + d} = \frac{3 + \frac{3}{4}}{1 + \frac{1}{4}} = \frac{15}{5} = 3.$

2. $f(x) = x + x\cos x; \quad x \geq 0$

$f(x) = x(1 + \cos x)$

$f(x) = 0 \Rightarrow x = 0$ and $\cos x = -1$

$\Rightarrow x = \pi, 3\pi, 5\pi \ldots$

$f(x) = x - x\cos x \ ; \ x < 0$

$= x(1 - \cos x)$

$f(x) = 0 \Rightarrow \cos x = 1; \quad x = 2\pi, 4\pi, 6\pi \ldots$

$\Rightarrow f(x) = 0$ for $x = 0, \pi, 2\pi, 3\pi, 4\pi \ldots$

$\Rightarrow f(x)$ is many-one

let's check for into OR onto

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$f(x)=\begin{cases}x(1+\cos x) & ;x\geq 0\\ x(1-\cos x) & ;x<0\end{cases}$

when, $x\geq 0$

$-1\leq\cos x\leq 1$

$0\leq 1+\cos x\leq 2$

$0\leq x(1+\cos x)\leq 2x$

$f(x)$ oscillates between 0 and $2x$

When $x\to\infty, f(x)\to\infty$

$\Rightarrow f(x)\in[0,\infty)$ *for* $x\geq 0.$

When $x<0$

$-1\leq\cos x\leq 1$

$0\leq 1+\cos x\leq 2$

$2x\leq x(1+\cos x)\leq 0$

as $x\to-\infty, f(x)\to-\infty$

$f(x)\in(-\infty,0)$ *for* $x<0;$

$\Rightarrow f(x)\in(-\infty,+\infty)\Rightarrow f(x)$ *is onto*

3. Area under the upper curve $g(x)=\sqrt{9-x^2}$

The equation $y=\sqrt{9-x^2}$ represents the upper half of a circle centered at the origin $(0,0)$ with radius $r=3$.

The area under this curve from $x=-3$ *to* $x=3$ is the area of a semicircle.

$\text{Area}_{\text{semicircle}}=\frac{1}{2}\pi r^2$

$=\frac{1}{2}\pi(3)^2$

$=\frac{9\pi}{2}$

Area under the lower curve $f(x)=\min(|x|+1,3-|x|)$

The function $f(x)$ is symmetric about the $y-axis$. We can calculate the area for $x\in[0,3]$ and and multiply by 2.

- Interval $[\mathbf{0},\ \mathbf{1}]: f(x)=x+1$. This forms a trapezoid with vertices $(0,\ 0),\ (1,\ 0),\ (1,\ 2),\ (0,\ 1)$.
- $\text{Area}_1=\frac{\text{sum of parallel sides}}{2}\times\text{width}=\frac{1+2}{2}\times 1=1.5$
- Interval $[\mathbf{1},\ \mathbf{3}]: f(x)=3-x$. This forms a **triangle** with vertices $(1,\ 0),\ (3,\ 0),\ (1,\ 2)$.
- $\text{Area}_2=\frac{1}{2}\times\text{base}\times\text{height}=\frac{1}{2}\times(3-1)\times 2=2$

Total area under $f(x)$ for $x\in[-3,3]$:

$\text{Area}_f=2\times(\text{Area}_1+\text{Area}_2)$

$=2\times(1.5+2)\ =2\times 3.5=7$

Final Area of Region S

The region S is the space trapped between the semicircle and the "tent-like" linear boundary $f(x)$.

$\text{Area}(S)=\text{Area}_{\text{semicircle}}-\text{Area}_f$

$=\frac{9\pi}{2}-7$

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4. Focus of Parabola $(k,\ 0)$.

Let say Q is (x_1, y_1)

Distance $d = x_1 + k = 2k$

$\Rightarrow x_1 = k$

Q on Parabola in 1st Quadrant

$\Rightarrow y_1^2 = 4k(k) \Rightarrow y_1 = 2k$

$\Rightarrow Q \equiv (k, 2k)$

Normal at Q:

$y^2 = 4kx$

$2y\frac{dy}{dx} = 4k$

$\frac{dy}{dx} = \frac{2K}{y}$

$\left.\frac{dy}{dx}\right|_Q = \frac{2K}{2K} = 1$

$m_{NQ} = -1$

$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x}{a^2}\frac{b^2}{2y} \Rightarrow \frac{dy}{dx} = -\frac{b^2}{a^2}\frac{x}{y}$

$\left.\frac{dy}{dx}\right)_Q = -\frac{b^2}{a^2}\frac{k}{2k}$

$\left.\frac{dy}{dx}\right)_Q = -\frac{b^2}{2a^2}$

m_{t_Q} (Ellipse) $= m_{N_Q}$ (Para) Given

$\frac{-b^2}{2a^2} = -1 \Rightarrow b^2 = 2a^2 \Rightarrow \frac{a^2}{b^2} = \frac{1}{2}$

$\Rightarrow b > a \quad (gmp)$

$a^2 = b^2(1 - e^2)$

$\frac{a^2}{b^2} = 1 - e^2$

$\Rightarrow e^2 = 1 - \frac{a^2}{b^2}$

$\Rightarrow e^2 = 1 - \frac{1}{2}$

$e = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$

5. $|M| = ad - bc$

$d = 0 / a = 0, \quad b = 0 / c = 0$

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$ad = 0$; either of them 0

$ad = 1$; both 1

Same $bc = 0$

$bc = 1$

$|M| = 0 - 0 = 0$

$|M| = 1 - 1 = 0$

$|M| = 1 - 0 = 1$

$|M| = 0 - 1 = -1$

$det(M)$ *can have 3 values*

$x = -1, \quad x = 0, \quad x = 1$

x	-1	0	$+1$
$P(x)$	$\frac{3}{16}$	$\frac{10}{16}$	$\frac{3}{16}$

$P(X = 1) = P(ad = 1 \text{ and } bc = 0) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$

For $ad = 1$ For $bc = 0$

0 0 0 1

1 0 1 0

0 1 0 0

1 1 1 1

$P(x = 0) = P(ad = 1 \text{ and } bc = 1) = \frac{1}{4} \cdot \frac{1}{4}$

OR

$P(ad = 0 \text{ and } bc = 0) = +\frac{3}{4} \cdot \frac{3}{4}$

$= \frac{10}{16}$

$P(x = -1) = P(ad = 0 \text{ and } bc = 1)$

$= \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}$

$t^2 - \gamma t + \delta = 0, \quad$ for non real roots

$\gamma^2 - 4\delta < 0$

$\gamma^2 < 4\delta$

$\left.\begin{matrix} \gamma & \delta \\ 0 & \neq 0 \neq -1 \\ -1 & \neq 0 \neq -1 \end{matrix}\right\} \Rightarrow \} \Rightarrow \begin{matrix} \delta = 1 \\ \gamma \in \{-1, 0, 1\} \end{matrix}$

$P(\text{non real}) = P(\delta = 1 \text{ and } \gamma = -1, 0, 1)$

OR $= P(\delta = 1)$ and $P(\gamma = -1)$

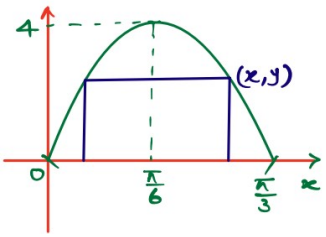
$P(\delta = 1)$ and $P(\gamma = 0) + P(\delta = 1)$ and $P(\gamma = 1)$

$= \frac{3}{16}\left(\frac{3}{16} + \frac{10}{16} + \frac{3}{16}\right)$

$= \frac{3}{16} = \frac{48}{256}$

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6.



$$y = 4\cos\left(\frac{\pi}{2} - 3x\right)$$

$$y = 4\sin 3x$$

$$0 \le x \le \frac{\pi}{3}$$

$$0 \le 3x \le \pi$$

$$Width \;\; w = 2\left|x - \frac{\pi}{6}\right| = \frac{\pi}{3} - 2x$$

$$Height \;\; h = 4\sin 3x$$

$$perimeter \;\; P = 2(w + h)$$

$$P = 2\left(\frac{\pi}{3} - 2x + 4\sin 3x\right)$$

$$\frac{dP}{dx} = 2(-2 + 4\cos 3x \cdot 3) = 0$$

$$12\cos 3x = 2$$

$$\cos 3x = \frac{1}{6}$$

$$\sin^2 3x + \cos^2 3x = 1$$

$$\sin^2 3x = 1 - \frac{1}{36} \Rightarrow \sin^2(3x) = \frac{35}{36}$$

$$\sin(3x) = \frac{\sqrt{35}}{6}$$

$$3x = \cos^{-1}\left(\frac{1}{6}\right)$$

$$A_0 = \left(\frac{\pi}{3} - 2x\right)4 \cdot \frac{\sqrt{35}}{6}$$

$$\cos^{-1}\left(\frac{1}{6}\right) + \sin^{-1}\left(\frac{1}{6}\right) = \frac{\pi}{2}$$

$$A_0 = \frac{2\sqrt{35}}{3}\left(\frac{\pi}{3} - \frac{\pi}{3} + \frac{2}{3}\sin^{-1}\left(\frac{1}{6}\right)\right)$$

$$3x = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{6}\right)$$

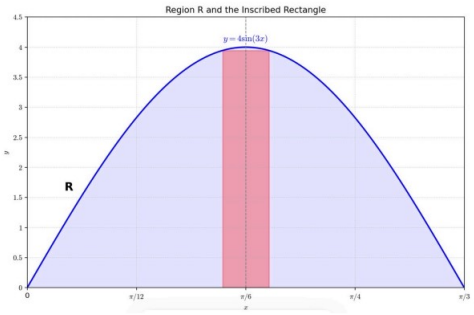
$$A_0 = \frac{2\sqrt{35}}{3}\left(\frac{2}{3}\sin^{-1}\left(\frac{1}{6}\right)\right)$$

$$x = \frac{\pi}{6} - \frac{1}{3}\sin^{-1}\left(\frac{1}{6}\right)$$

$$2x = \frac{\pi}{3} - \frac{2}{3}\sin^{-1}\left(\frac{1}{6}\right)$$

$$\frac{9A_0}{\sqrt{35}\sin^{-1}\left(\frac{1}{6}\right)} = 4.$$

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7. $\dfrac{(\sec\theta+\csc\theta+\cot\theta)^{2011}}{2011}+C$

$P=2011; f(\theta)=\sec\theta+\csc\theta+\cot\theta$

$f(\pi/3)=2+\dfrac{2}{\sqrt{3}}+\dfrac{1}{\sqrt{3}}$

$=\dfrac{2\sqrt{3}+3}{\sqrt{3}}=2+\sqrt{3}$

$f(\pi/4)=\sqrt{2}+\sqrt{2}+1=\sqrt{8}+1$

8. $|A|\neq 0,\ given;$

$A\neq I\ \ given$

$A^2=A^T$

$|A^2|=|A^T$

$|A|^2=|A|$

$|A|^2-|A|=0\Leftrightarrow|A|(|A|-1)=0$

$|A|=0;\ \ N.P\,|A|=1$

$\Rightarrow A$ is Invertible

$A^2=A^T$

$(A^2)^T=(A^T)^T=A$

$\Rightarrow(A^2)^T=A\Rightarrow(A^T)^2=A$

$\Rightarrow(A^2)^2=A\Rightarrow A^4=A$

$A^4-A=0 \qquad OR \qquad A^4=A$

$A(A^3-I)=0 \qquad\qquad A\ isi\ \ nvertible$

$\Rightarrow|A|\neq 0\Rightarrow A^3=I \qquad A^3=I$

$A^3-I=0\Rightarrow A^3-0A^2-0A-I=0$

$(A-I)(A^2+A+I)=0 \ \ \Rightarrow tr(A)=0$

$Now\,|A-I||A^2+A+I|=0$

$atleast\ one\ of\ \ them\ must\ be\ \ zero$

$(1)\,|A^2+A+I|=0$

$(2)\,|A-I|=0$

$let\,|A^2+A+I|\neq 0$

$\Rightarrow A^2+A+I\ is\ \ Invertible\ (A-I)(A^2+A+I)(A^2+A+I)^{-1}=0$

$(A-I)I=0$

$A-I=0$

$A=I$ (contradiction given $A\neq I$)

$\Rightarrow\left|A^2+A+I\right|=0$

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9. Translate conditions into algebraic equations

Let $z = x + iy$, where $x, y \in \mathbb{R}$.

From condition **1**: $|z - 1| = 2$

$|x + iy - 1| = 2$

$(x-1)^2 + y^2 = 4 \quad \ldots(1)$

From condition $2: |z^2 - 1| = 6$

Note that $z^2 - 1 = (z-1)(z+1)$. Thus:

$|(z-1)(z+1)| = 6$

$|z-1| \cdot |z+1| = 6$

Substitute $|z-1| = 2$ into the equation:

$2 \cdot |z+1| = 6$

$|z+1| = 3$

$(x+1)^2 + y^2 = 9 \quad \ldots(2)$

Solve for x and y

Subtract equation (1) from equation (2):

$[(x+1)^2 + y^2] - [(x-1)^2 + y^2] = 9 - 4$

$(x^2 + 2x + 1) - (x^2 - 2x + 1) = 5$

$4x = 5$

$x = \frac{5}{4} = 1.25$

Now, substitute $x = \frac{5}{4}$ back into equation (1) to find y^2 :

$\left(\frac{5}{4} - 1\right)^2 + y^2 = 4$

$\left(\frac{1}{4}\right)^2 + y^2 = 4$

$\frac{1}{16} + y^2 = 4$

$y^2 = \frac{63}{16}$

Taking the square root:

$y = \pm\frac{\sqrt{63}}{4} = \pm\frac{3\sqrt{7}}{4}$

Evaluate the options

Option (A): Calculate $\left|\frac{\text{Im}(z)}{\text{Re}(z)}\right| \cdot \sqrt{7}$

$\left|\frac{y}{x}\right| \cdot \sqrt{7} = \frac{\frac{3\sqrt{7}}{4}}{\frac{5}{4}} \cdot \sqrt{7}$

$= \frac{3\sqrt{7}}{5} \cdot \sqrt{7}$

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$= \frac{3 \cdot 7}{5} = \frac{21}{5} = 4.2$

Option (B) : $\mathrm{Re}(z) = x = 1.25$. *This is correct.*

Option(C) : $[\mathrm{Im}(z)]^2 = y^2 = \frac{63}{16}$.

Option (D) : $|z| = \sqrt{x^2 + y^2}$

$|z|^2 = \left(\frac{5}{4}\right)^2 + \frac{63}{16}$

$= \frac{25}{16} + \frac{63}{16}$

$= \frac{88}{16} = \frac{11}{2} = 5.5$

$|z| = \sqrt{5.5} \neq \sqrt{7}$

10. When $n \in a_n$ then $n-1 \notin a_n \Rightarrow a_{n-2}$

When $n \notin a_n$ then $n-1 \in a_n \Rightarrow a_{n-1}$

$\Rightarrow a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$

No. of subset when not to select two consecutive

$\{1\} = 2 \rightarrow \varnothing, \{1\}$

$\{1,2\} = 3 \rightarrow \varnothing, \{1\}, \{2\}$

$\{1,2,3\} = 5 \rightarrow \varnothing, \{1\}, \{2\}, \{3\}, \{1,3\}$

$a_3 = a_2 + a_1 = 3 + 2$

$a_4 = a_3 + a_2 = 5 + 3 = 8$

$a_5 = a_4 + a_3 = 8 + 5 = 13$

$a_6 = a_5 + a_4 = 13 + 8 = 21$

11. The plane P_1 passes through the intersection of $x - y - z - 4 = 0$ and $x + y + 2z - 4 = 0$ its equation is:

$(x - y - z - 4) + \lambda(x + y + 2z - 4) = 0$

Grouping terms by variables:

$(1 + \lambda)x + (\lambda - 1)y + (2\lambda - 1)z - 4(1 + \lambda) = 0$

The normal vector to P_1 is $\vec{n}_1 = (1 + \lambda, \lambda - 1, 2\lambda - 1)$.

P_1 is parallel to the line of intersection of $2x + 3y + z = 1$ *and* $x + 3y + 2z = 2$. The direction vector

v of this line is the cross product of the normals of these two planes:

$$\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{vmatrix}$$

$= \hat{i}(6-3) - \hat{j}(4-1) + \hat{k}(6-3)$

$= 3\hat{i} - 3\hat{j} + 3\hat{k} \propto \hat{i} - \hat{j} + \hat{k}$

Since the plane is parallel to the line, the normal n_1 must be perpendicular to $v\,(n_1 \cdot v = 0)$:

$(1 + \lambda)(1) + (\lambda - 1)(-1) + (2\lambda - 1)(1) = 0$

$1 + \lambda - \lambda + 1 + 2\lambda - 1 = 0$

$2\lambda + 1 = 0 \Rightarrow \lambda = -1/2$

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Substitute $\lambda = -1/2$ back into the equation from Step1.1:

$\frac{1}{2}x - \frac{3}{2}y - 2z - 2 = 0 \Rightarrow x - 3y - 4z - 4 = 0$

Comparing this to $x + \alpha y + \beta z + \gamma = 0$, we get $\alpha = -3, \beta = -4, \gamma = -4$.

Thus, $|\alpha + \beta + \gamma| = |-3 - 4 - 4| = 11$. (Statement A is correct).

The normal vector is $(1, -3, -4)$, which is not parallel to $(2, -1, -3)$. (Statement C is in correct).

To find the area of the quadrilateral $V_1V_2V_3V_4$ using the projection method, we first define a reference quadrilateral in space using points from the four lines and then project its area vector onto the direction of the lines.

Define the Reference Quadrilateral

We select one point from each line L_1, L_2, L_3, L_4:

$Q_1 = (3,3,3)$ *from* L_1

$Q_2 = (3,3,0)$ *from* L_2

$Q_3 = (0,3,0)$ *from* L_3

$Q_4 = (0,3,3)$ *from* L_4

These points form a rectangle in the plane $y = 3$. We calculate the area vector A of this rectangle:

$\overrightarrow{Q_2Q_3} = (0-3, 3-3, 0-0) = (-3,0,0)$

$\overrightarrow{Q_2Q_1} = (3-3, 3-3, 3-0) = (0,0,3)$

The area vector $\vec{A}$ is the cross product of these two adjacent sides:

$\vec{A} = \overrightarrow{Q_2Q_3} \times \overrightarrow{Q_2Q_1}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 0 & 0 \\ 0 & 0 & 3 \end{vmatrix}$$

$= \hat{i}(0) - \hat{j}(-9) + \hat{k}(0) = 9\hat{j}$

The magnitude of this area is $|\vec{A}| = 9$, and its normal is $\hat{n} = (0,1,0)$.

The four lines are parallel to the vector:

$\vec{u} = 2\hat{i} + \hat{j} + 2\hat{k}$

The unit vector in this direction is:

$\hat{u} = \frac{2\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{1}{3}(2,1,2)$

Calculate the Minimum Area S

The minimum area S is the area of the cross-section perpendicular to the lines. Geometrically, if a plane has an area vector A, its projection onto a plane with unit normal $\hat{u}$ is given by the dot product:

$S = |\vec{A} \cdot \hat{u}|$

Substituting our values:

$S = |(0,9,0) \cdot \frac{1}{3}(2,1,2)|$

$= |\frac{1}{3}(0 \cdot 2 + 9 \cdot 1 + 0 \cdot 2)|$

$\frac{9}{3}$

$= 3$

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12. $\int_0^1 \left(\ln(x) \sum_{n=0}^{\infty} x^{2n} \right) dx$

$= \sum_{n=0}^{\infty} \int_0^1 \ln x \cdot x^{2n} dx$

Using integration by parts $\int_0^1 x^m \ln(x) dx = -\frac{1}{(m+1)^2}$

We know the sum reciprocals of squares is $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. The sum of odd squares is:

$$\sum_{n=\text{odd}} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=\text{even}} \frac{1}{n^2} = \frac{\pi^2}{6} - \frac{1}{4}\left(\frac{\pi^2}{6}\right) = \frac{\pi^2}{8}$$

Comparing $-\frac{\pi^2}{8}$ with $-\frac{\pi^2}{\lambda}$ we find

$\lambda = 8$

$\alpha = 2, \beta = 6$

13. $x\frac{dy}{dx} + y = \frac{y^2 \ln x}{x}$

Divide by xy^2

$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{\ln x}{x^2}$

$let \frac{1}{y} = t \; ; \; -\frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$

$-\frac{dt}{dx} + \frac{1}{x}t = \frac{\ln x}{x^2}$

$\frac{dt}{dx} - \frac{t}{x} = \frac{-\ln x}{x^2}$ (L.D.E)

$I.F = e^{\int -\frac{1}{x} dx}$

$= e^{\ln(\frac{1}{x})}$

$= \frac{1}{x}$

$t \cdot \frac{1}{x} = \int \frac{-\ln x}{x^2} \cdot \frac{1}{x} dx \rightarrow (1)$

$= \int \frac{-\ln x}{x^3} dx$

Let say $I_1 = \int \ln x \cdot x^{-3} dx$

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$$= \ln x \frac{x^{-2}}{-2} - \int \frac{1}{x} \frac{x^{-2}}{-2} dx$$

$$= -\frac{\ln x}{2} \frac{1}{x^2} - \frac{x^{-2}}{4}$$

$$= \frac{-\ln x}{2x^2} - \frac{1}{4x^2}$$

$$t = \frac{1}{y} = \frac{1}{f(x)}$$

$$\frac{1}{f(x) \cdot x} = \frac{\ln x}{2x^2} + \frac{1}{4x^2} + C$$

$$\frac{1}{f(x) \cdot x} = \frac{2\ln x + 1}{4x^2} + C$$

$$, \overset{x=1}{f}(1) = 1 : \frac{1}{1} = \frac{2\ln 1 + 1}{4} + C$$

$$1 = \frac{1}{4} + c$$

$$c = \frac{3}{4}$$

$$\frac{1}{f(x) \cdot x} = \frac{2\ln x + 1}{4x^2} + \frac{3}{4}$$

$$I = \int_1^e \frac{1}{x^2 f(x)} dx$$

$$I = \int_1^e \left(\frac{2\ln x + 1}{4x^3} + \frac{3}{4x} \right) dx$$

$$I = \int_1^e \frac{\ln x}{2x^3} dx + \int_1^e \frac{1}{4x^3} dx + \frac{3}{4} \int_1^e \frac{1}{x} dx$$

$$= \frac{1}{2} \left(\ln x \left(\frac{x^{-2}}{-2} \right) - \int \frac{1}{x} \left(\frac{x^{-2}}{-2} \right) dx \right)$$

$$= \frac{-1}{4} \frac{\ln x}{x^2} - \frac{x^{-2}}{8} + \frac{1}{4} \frac{x^{-2}}{-2} + \frac{3}{4} \ln x$$

$$= \left(-\frac{1}{4} \frac{\ln x}{x^2} - \frac{1}{4x^2} + \frac{3}{4} \ln x \right)_1^e$$

$$= \frac{-1}{4e^2} - \frac{1}{4e^2} + \frac{3}{4} + \frac{1}{4}$$

$$= \frac{-2}{4e^2} + 1$$

$$I = \frac{-1}{2e^2} + 1$$

$$\frac{+1}{2e^2} = (1 - I)$$

$$\frac{4e^2}{2e^2} = (1 - I)4e^2$$

$$2 = (1 - I)4e^2$$

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14. $a_2 = 1+d, \quad a_5 = 1+4d, \quad a_{14} = 1+13d$

$(1+4d)^2 = (1+d)(1+13d)$

$1+16d^2+8d = 1+13d+d+13d^2$

$3d^2 - 6d = 0$

$3d(d-2) = 0$

$d = 0$ (*Not possible*) $\Rightarrow d = 2$

$S_n = \frac{n}{2}[2\cdot 1+(n-1)2]$

$S_n \Rightarrow n+n^2-n = \mathbf{n^2}$

$\frac{a_5}{a_2} = \frac{1+4\cdot 2}{1+2} = \frac{9}{3} = \mathbf{3}$

$T_m = \frac{1(3^m-1)}{3-1}$

$n^2 = \frac{3^m-1}{2} \Rightarrow 2n^2 = 3^m-1$

$\Rightarrow \frac{3^m-1}{2} = n^2$

$m = 1 : 1 \Rightarrow n = 1$ (*given* $n > 1$)

$m = 2 : \frac{9-1}{2} = 4 \Rightarrow n = 2$

$m = 3 : \frac{27-1}{2} = 13 \Rightarrow n \notin \mathrm{I}$

$m = 4 : \frac{81-1}{2} = 40 \Rightarrow n \notin \mathrm{I}$

$m = 5 : \frac{243-1}{2} = 121 \Rightarrow n = 11$

$m = 6 : \frac{729-1}{2} = n \notin \mathrm{I}$

$(n,m) = (2,2) \quad (11,5)$ 2 Ordered pair

15. Let $S = 2\sin 2^\circ + 4\sin 4^\circ + \ldots + 176\sin 176^\circ + 178\sin 178^\circ$

By reversing the order of terms and using the identity $\sin\theta = \sin(180^\circ - \theta)$:

$S = 178\sin 2^\circ + 176\sin 4^\circ + \ldots + 4\sin 176^\circ + 2\sin 178^\circ$

Adding the two equations together:

$2S = 180(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \ldots + \sin 176^\circ + \sin 178^\circ)$

Dividing by 2 and writing in sigma notation:

$S = 90\sum_{k=1}^{89} \sin 2k^\circ$

$$\sin\alpha + \sin(\alpha+\beta) + \ldots + \sin(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \sin\left(\alpha+(n-1)\frac{\beta}{2}\right)$$

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$$S=90\frac{\sin\left(89\cdot\frac{2^\circ}{2}\right)}{\sin\left(\frac{2^\circ}{2}\right)}\sin\left(2^\circ+(89-1)\frac{2^\circ}{2}\right)$$

$$S=90\frac{\sin(89^\circ)}{\sin(1^\circ)}\sin(90^\circ)$$

$$s=90\frac{\cos 1^\circ}{\sin 1^\circ}\Rightarrow\frac{s}{\cot 1^\circ}=90$$

16. Since the vectors $\vec{u},\vec{v}$ and $\vec{w}$ form the sides of triangle ABC

$\vec{u}+\vec{v}+\vec{w}=\vec{0}\Rightarrow\vec{w}=-\left(\vec{u}+\vec{v}\right)$

$\vec{w}=-\vec{u}-\vec{v}$ in to expression:

.Numerator(N):

$N=\vec{u}.\left(\vec{w}-\vec{v}\right)$

$=\vec{u}.\left(-\vec{u}-\vec{v}-\vec{v}\right)=\vec{u}.\left(-\vec{u}-2\vec{v}\right)$

$=\left|\vec{u}\right|^2-2\left(\vec{u}.\vec{v}\right)$

.Denominator(D):

$D=\vec{w}.\left(\vec{u}-\vec{v}\right)$

$=\left(-\vec{u}-\vec{v}\right).\left(\vec{u}-\vec{v}\right)=\left(\left|\vec{u}\right|^2-\left|\vec{v}\right|^2\right)$

$=\left|\vec{v}\right|^2-\left|\vec{u}\right|^2$

Dot product $\vec{u}.\vec{v}$

Substitute the simplified N and D back into the given equation:

$$\frac{\vec{u}.\left(\vec{w}-\vec{v}\right)}{\vec{w}.\left(\vec{u}-\vec{v}\right)}=\frac{2}{5}$$

Substitute the simplified magnitudes $\left|\vec{u}\right|$=4 and $\left|\vec{v}\right|$=6:

$$\frac{-4^2-2\left(\vec{u}.\vec{v}\right)}{6^2-4^2}=\frac{2}{5}\Rightarrow\quad\frac{-16-2\left(\vec{u}.\vec{v}\right)}{36-16}=\frac{2}{5}$$

$$\frac{-16-2\left(\vec{u}.\vec{v}\right)}{20}=\frac{2}{5}\Rightarrow\quad-16-\left(\vec{u}.\vec{v}\right)=\frac{2}{5}\times 20$$

$$-16-2\left(\vec{u}.\vec{v}\right)=8\Rightarrow\quad-2\left(\vec{u}.\vec{v}\right)=24$$

$\vec{u}.\vec{v}=-12$

Calculate $\left|\vec{u}\times\vec{v}\right|^2$ Using Lagrange's Identity

The identity states:

$\left|\vec{u}\times\vec{v}\right|^2=\left|\vec{u}\right|^2\left|\vec{v}\right|^2-\left(\vec{u}.\vec{v}\right)^2$

$\left|\vec{u}\times\vec{v}\right|^2=\left(4^2\right)\left(6^2\right)-\left(-12\right)^2$

$=(16)(36)-144=576-144$

$=432$

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17. $x \in I \Rightarrow \{x\} = 0 \Rightarrow \lambda_1 = 0$

$t = \log_{[x]}\{x\}$

$t^2 - 3t + 2 = 0$

$t = 2, \quad t = 1$

$\log_{[x]}\{x\} = 2; \quad \{x\} = [x]$

N.P.

$\{x\} = [x]^2$

$N.P$

18. Let $t = x \ln x$

As $x \to 0^+$, *we know*:

$t \to 0$

Also, $x^x = e^{x \ln x} = e^t$

$\ln(x^x) = t, \quad x^x - 1 = e^t - 1$

So the expression becomes:

$$L = \lim_{t \to 0} \frac{t(e^{e^t - 1} - 1) - (e^t - 1)}{\sin^k t}$$

Expansion of $e^t - 1$:

$$e^t - 1 = t + \frac{t^2}{2} + \frac{t^3}{6} + \ldots$$

Expansion of $e^{e^t - 1} - 1$

Let $u = e^t - 1 = t + \frac{t^2}{2} + \frac{t^3}{6} + \ldots$

Then:

$$e^u - 1 = u + \frac{u^2}{2} + \frac{u^3}{6} + \ldots$$

Substitute u:

$$e^{e^t - 1} - 1 = t + t^2 + \frac{5}{6}t^3 + O(t^4)$$

Evaluate Numerator

$N = t(e^{e^t - 1} - 1) - (e^t - 1)$

Substitute expansions:

$$N = t\left(t + t^2 + \frac{5}{6}t^3\right) - \left(t + \frac{t^2}{2} + \frac{t^3}{6}\right)$$

$$N = -t + \frac{1}{2}t^2 + \frac{5}{6}t^3 + O(t^4)$$

Denominator

$\sin^k t \sim t^k$

Determine k

For the limit to be **finite and non-zero**, powers must match.

Leading term in numerator:

$N \sim -t$

So:

$t^k \sim t \Rightarrow k = 1$

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PHYSICS

19. Use the property of cental forces in circular motion. For a force directed toward a point on the circumference, the force law follows $F \propto \frac{1}{r^5}$ to satisfy the orbit geometry.

20. The maximum angle of the cone is the angle of repose θ, where $\tan\theta = \mu$. Volume $V = \frac{1}{3}\pi R^2 h$. Express R and h in terms of $C = 2\pi R$ and μ.

21. Use the constraint relation. The acceleration of the string end being pulled is a_p, and the small body's acceleration is a. Because of the halfway wrap, the acceleration of the center of the disc also plays a role.

22. Calculate the horizontal and vertical components of the hydrostatic force on the quarter-cylinder. Use torque equilibrium about the hinge C to find the cable tension.

23. Treat this as a combination of two lenses. Find the focal length of each part and determine the position where the divergent or convergent rays produce the minimum spot size (circle of least confusion).

24. In a time-varying magnetic field, the induced electric field is non-conservative. The voltmeter reading depends on the path of the leads because the line integral $\oint \vec{E}.d\vec{l}$ is non-zero.

25. For the cone to roll "up," its center of mass must move down. This depends on the divergence of the rails (α) relative to the cone's semi-vertical angle (β).

26. Use the definition of the center of mass (COM) for a spherical surface. A narrow lune behaves like a semi-circular ring for very small θ.

27. Apply Wien's displacement law $\left(T \propto \frac{1}{\lambda_{max}}\right)$ and the Stefan-Boltzmann Law $\left(P = \sigma S T^4\right)$. Identical luminosity with different temperatures implies significantly different radii.

28. Use the propagation of errors for $R = \frac{Ve^L}{nT}$. Note that the relative error in P is equal to the absolute error in $L\left(\Delta P / P = \Delta L\right)$

29. This is cycloidal motion. The particle's trajectory is the result of the constant electric force and the velocity-dependent magnetic force.

30.

Velocity of wave in rope = $\sqrt{T/\mu} = \sqrt{\frac{8}{20\times10^{-3}}} = 20 m/s$

(a) Wavelength = $\frac{v}{f} = \frac{20}{100} m = 20$ cm

(b) Max. transverse acceleration = $\omega^2 A$

$= (2\pi \times 100)^2 \times \frac{2}{100} = 800\,\pi^2$ m/s^2

(c) Equation of wave = y = A sin ($\omega t - kx + \phi$) = 0.02 sin ($200\,\pi - \frac{2\pi}{0.2}x + \phi$)

at t = 0, x = 0 y = – A

$-A = A\sin\phi$

$\phi = -90°$

so equation is : $y = -0.02 \cos(200\,\pi - 10\,\pi x)$

(d) if f = 200 Hz $\lambda = \frac{v}{f} = \frac{20}{200} = 10$cm

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31. Sum the Laplace pressure force (ΔP . Area) and the surface tension pull (T.Perimeter). Given $R_1 = 6R_2$, the pressure difference is $T\left(\frac{1}{R_2} - \frac{1}{R_1}\right)$.

32. The induced dipole is $p = \alpha E$. The force is $p(dE/dr)$. Substituting $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ yields $F \propto \frac{1}{r^5}$.

33. Use the polytropic molar heat capacity formula $C = C_\upsilon + \frac{R}{1-n}$. For a monoatomic gas, $C_\upsilon = \frac{3}{2}R$. Determine *n* and then calculate the ratio W/Q.

34. Conceptual

35. The maximum field E_{max} is at the inner radius *a*. Relate V_{max} to E_{max} using $V = \int E dr$. Then use the energy formula $U = \frac{1}{2}C'V_{max}^2$.

36.

$\lambda = 600$ nm $I_{max} = 4I_0$ $\beta = \frac{\lambda}{d} = 0.1°$

$I = 4I_0\cos^2\frac{\theta}{2}$

$2I_0 = 4I_0\cos^2\frac{\theta}{2}$

$\cos\frac{\theta}{2} = \pm\frac{1}{\sqrt{2}}$

$\therefore \theta = \frac{\pi}{2} = \frac{2\pi}{\lambda}\Delta x$

$\Delta x = \frac{\lambda}{4}$

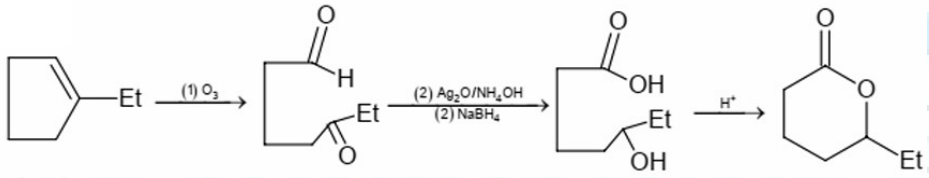
$(\mu - 1)t = \frac{\lambda}{4}$

$\mu - 1 = \frac{1}{4}\times\frac{600\times10^{-9}}{250\times10^{-9}} = 0.6$

$\therefore \mu = 1.6$

CHEMISTRY

37.



39.

$n_{Hx} = n_{NaOH} = 0.25 \times V_{NaOH}$, pH = pK_a = 5 (acid buffer)

$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C = 9$ $\Rightarrow \log C = -1 \Rightarrow C = 10^{-1}$

$C = \frac{0.25V}{60 + V_{NaOH}} \Rightarrow V_{NaOH} = 40$ ml

n_{Hx} = 10 mm = 0.82 gm

$\% = \frac{0.82}{1.025}\times100 = 80\%$

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41.

$$MnO_4^- + Fe^{+2} + 8H^{\oplus} \longrightarrow Mn^{+2} + Fe^{+3}$$

0.1 M

0.01M 0.09 M

$$E^0_{R.P\left(MnO_4^{-2}/Mn^{+2}\right)} > E^0_{R.P\left(H^+/H_2\right)}$$

So permanganate electrode behave as cathode.

$E_{cell} = 1.51 - 0 = 1.51.$

Using Nernst's equation :- (n = 5)

$$E_{MnO_4^-/Mn^{+2}} = 1.51 + \frac{0.059}{5}\log\frac{[MnO_4^-][H^+]^8}{[Mn^{+2}]}$$

$= 1.51 + 0.11 = 1.4$ V

42.

$$\underset{\text{(Anion)}}{\underset{X}{NO_3^-}} + H_2SO_4 \underset{\text{(Conc.)}}{} \rightarrow \underset{\text{Brown fumes}}{NO_2\uparrow} + H_2O$$

$$\underset{Sol^n}{FeSO_4} + \underset{conc.}{H_2SO_4} + \underset{X}{NO_3^-}$$

$\downarrow$

$[Fe(H_2O)_5(NO)]SO_4$

(Dark brown ring)

$$\underset{\text{(cation)}}{\underset{X}{Cu^{2+}}} + (\text{dil } HCl + H_2S)$$

(Group-II reagent)

$\downarrow$

$CuS\downarrow$

(Black ppt)

(Y)

$$\underset{(Y)}{CuS} \xrightarrow[HNO_3]{Conc^n} \underset{\text{Soluble}}{Cu(NO_3)_2} + NO_2 + S + H_2O$$

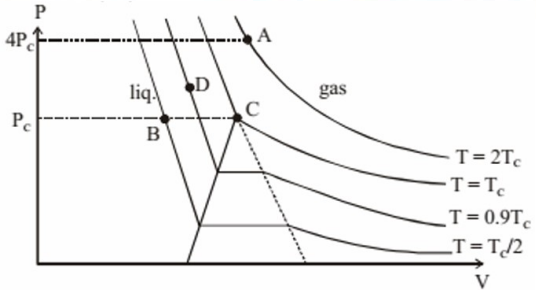
$\downarrow$ Excess NH_4OH Sol^n.

$[Cu(NH_3)_4]^{2+}$

Deep blue colour solution.

$\therefore X \rightarrow Cu(NO_3)_2$

43.

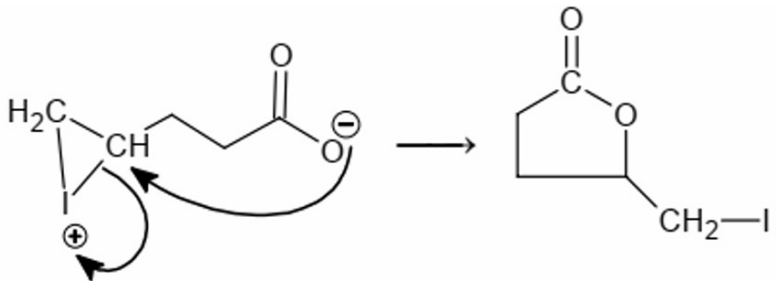


46. Both complexes have $Cr^{+3}\left(d^3\right)$, and coordination number is 6. Hence d^2sp^3 $Pd^{+2}\left(d^8\right)$ complexes with coordination number $\rightarrow 4$ are square plannar and magnetic moment is zero

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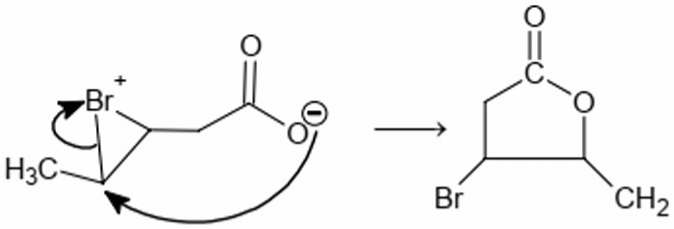
48.

(A)

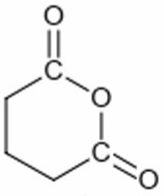


I_2 will attack the double bond to form a cyclic halonium ion. It then undergoes on intramolecular reaction. Formation of five membered ring is favoured because (i) attack by a nucleophile on a halonium ion normally takes place at the more substituted site.
(ii) Formation of five membered ring is more probable.

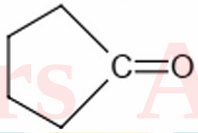
(B)



(C)

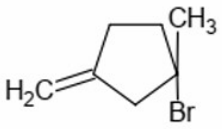
Option(C)gives 

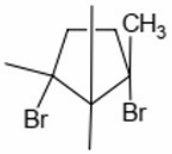
(D)

Option(D)gives 

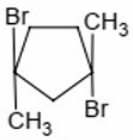
1, 4 and 1,5-dicarboxylic acids give anhydrides whereas 1,6 and 1, 7-diacids give ketones when heated with P_2O_5.

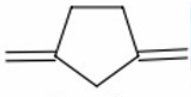
51.

A = 

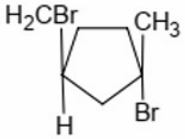
B = 

Inactive

C = 

D = 

Inactive

E = 

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52.

$$t_{1/2} \propto \frac{1}{a^{1-n}}$$

$\Rightarrow t_{1/2} = K a^{1-n}$

K^{-1} rate constant

$\text{Log } t_{1/2} = \log K + (1 - n) \log a$

$Y = C + mx$

Slope $= (1 - n) = -2$

$\Rightarrow n = 3$

53.

$$CO(g) + Cl_2(g) \underset{}{\overset{K=2\times10^7}{\rightleftharpoons}} COCl_2(g)$$

$$Q_P = \frac{P^1_{COCl_2}}{P^1_{CO} \cdot P^1_{Cl_2}} = \frac{1/4}{1/4 \times 2/4} = 2$$

$\Delta_r G = \Delta_r G^\circ + 2.303\ RT \log Q$

$= 2.303\ RT \log Q/K$

$= 2.303 \times 8.3 \times 300 \times (-7)$

$= -40.141$ kJ/mole

$\simeq -40.14$ kJ/mole

54.

$A \rightarrow 2B + C$

$2 - x \quad 2x \quad x$

At 12 hrs

$$20 = \frac{10}{10 + 2 + 2x} \times 26$$

$12 + 2x = 13$

$x = \frac{1}{2}$ at 24 hrs

$$\frac{2}{2 - 0.5} = \frac{2 - 0.5}{A_t}$$

$A_t = 1.125, x = 0.875$

$$P_{H_2O} = \frac{10}{10 + 2 + 2 \times 0.875} \times 26 = 18.909\ mmHg$$

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